Practical - 9

Practical Title: Secure User Registration with Database Insert, Duplicate Email Check, and Password Hashing

Problem Definition: Implement secure user registration by inserting user details into MySQL using MYSQLi. Validate input on frontend and backend, check duplicate email/username, hash passwords using password_hash(), and display appropriate messages.

A. Theory

1. Is password_hash() used before storing passwords?

Answer: Yes, password_hash() is used to generate a secure password hash before storing the password in the database. This prevents passwords from being stored as plain text.

2. Is duplicate email/username checked?

Answer: Yes, the database is checked for existing email addresses before inserting a new user. This helps prevent duplicate accounts. Unique constraints in the database provide additional protection against duplicates.

3.Are frontend and backend validations consistent?

Answer: Yes, frontend validation provides immediate feedback to users, while backend validation checks the submitted data again before processing it. Both validations should follow the same rules to ensure data consistency and sec-urity.

B. Implementation

Storing passwords

```
$passwordHash =  password_hash( $password, PASSWORD_DEFAULT);
```

Checking for duplicates

```
$checkQuery = "

    SELECT student_id

    FROM students
```

```
    WHERE studentid = :studentid

       OR email = :email

    LIMIT 1

";

$checkStmt = $pdo->prepare($checkQuery);

$checkStmt->execute([

    ":studentid" => $studentid,

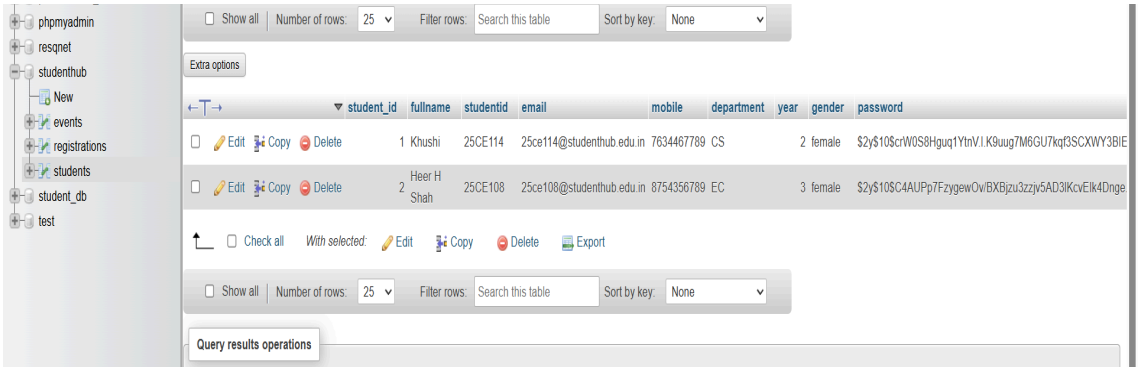
    ":email" => $email

]);

$existingStudent = $checkStmt->fetch();
```

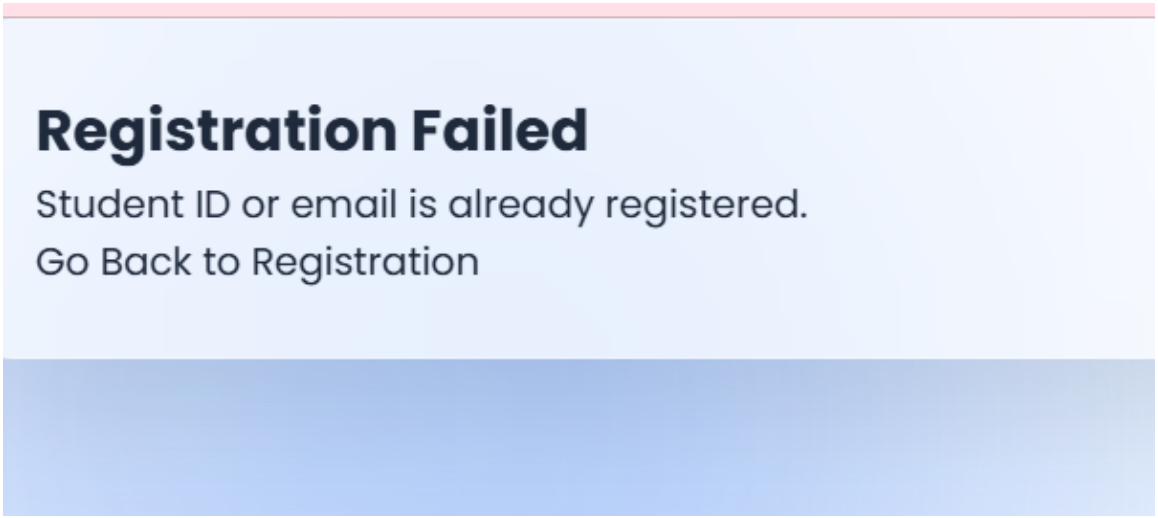
C. Evidence of Work (Screenshots)

Figure 1.1 Database in phpmyadminn



Description: Passwords are stored in hash

Figure 1.2 Message showing



Description: When we try registering duplicate email, this message shows and registration fails

D. GitHub Repository Link:

https://github.com/heerce27/StudentHub-Portal.git

E. Challenges Faced:

Checking for duplicate email addresses and usernames before inserting new users into the database.

F. Learning Outcomes:

- Learned how to implement secure user registration using PHP and MySQLi.
- Learned to check duplicate email addresses and usernames before inserting records.
- Understood how to use password_hash() to securely hash passwords before storing them in the database.

Faculty Remarks: